

Practical 2 Tasks

HTTP Protocol Exercises

1. Capture HTTP packets while opening a website in a browser and identify:
 - Request method
 - Response code
 - Content type
 - Host field
2. Compare HTTP GET and POST requests using packet captures.
3. Find the HTTP response time for loading a webpage.
4. Identify all objects transferred during a webpage load (HTML, CSS, images, JavaScript).
5. Capture packets while downloading a file and determine:
 - File size
 - MIME type
 - Transfer duration
6. Analyze HTTP status codes such as:
 - 200 OK
 - 301 Redirect
 - 404 Not Found
 - 500 Internal Server Error
7. Use Wireshark filters to display only HTTP traffic.
8. Identify persistent and non-persistent HTTP connections from packet captures.
9. Observe cookies exchanged between browser and server.
10. Analyze HTTP headers and explain the purpose of:
 - User-Agent
 - Accept-Language
 - Cache-Control
 - Content-Length

DNS Protocol Exercises

1. Capture DNS query and response packets when accessing a website.
2. Identify:
 - Query type
 - Domain name
 - IP address returned
3. Compare recursive and iterative DNS query behavior.
4. Find the Time-To-Live (TTL) value in DNS responses.
5. Analyze DNS traffic for multiple websites and compare response times.
6. Use Wireshark filters to display only DNS traffic.
7. Identify the DNS server used by the system.
8. Capture reverse DNS lookup traffic and explain its purpose.
9. Observe DNS packet structure including:
 - Header
 - Question section
 - Answer section

10. Compare UDP-based and TCP-based DNS communication.

TCP Protocol Exercises

1. Capture a TCP three-way handshake and identify:
 - SYN
 - SYN-ACK
 - ACK packets
2. Analyze TCP sequence numbers and acknowledgment numbers.
3. Identify TCP flags in captured packets:
 - SYN
 - ACK
 - FIN
 - RST
 - PSH
4. Measure TCP connection setup time.
5. Observe TCP connection termination using FIN packets.
6. Compare packet retransmissions during network congestion.
7. Use Wireshark to analyze TCP flow control and window size.
8. Identify duplicate ACK packets and explain their significance.
9. Capture file transfer traffic and study TCP throughput.
10. Compare TCP performance for small and large file transfers.
11. Analyze TCP checksum fields and error detection.
12. Use Wireshark statistics tools to examine TCP conversations.

UDP Protocol Exercises

1. Capture UDP packets generated by DNS or streaming applications.
2. Compare TCP and UDP packet headers.
3. Identify source and destination ports in UDP communication.
4. Measure delay variation in UDP traffic.
5. Analyze UDP traffic generated by online video or voice applications.
6. Use Wireshark filters to display only UDP traffic.
7. Observe packet loss behavior in UDP communication.
8. Compare reliability mechanisms in TCP and UDP using captured traffic.
9. Analyze packet size variations in UDP applications.
10. Study real-time traffic behavior using UDP streams.

NAT (Network Address Translation) Exercises

1. Capture packets before and after NAT translation and identify IP changes.
2. Analyze source NAT and destination NAT behavior.
3. Observe private and public IP address mapping.
4. Identify port translation performed by NAT devices.

5. Compare static NAT and dynamic NAT operations.
6. Use Wireshark to examine packets passing through a home router.
7. Analyze how NAT affects TCP and UDP communication.
8. Observe packet headers modified by NAT devices.
9. Study internet access from multiple devices behind a NAT router.
10. Explain the role of NAT in IPv4 address conservation.

ICMP Protocol Exercises

1. Capture ICMP packets generated using the ping command.
2. Identify:
 - Echo Request
 - Echo Reply packets
3. Measure round-trip time (RTT) using ICMP packets.
4. Use traceroute or tracert and analyze ICMP Time Exceeded messages.
5. Compare ICMP traffic for reachable and unreachable hosts.
6. Analyze Destination Unreachable ICMP messages.
7. Use Wireshark filters to display only ICMP traffic.
8. Observe TTL values in ICMP packets.
9. Capture fragmented ICMP packets and analyze fragmentation behavior.
10. Study network diagnostics using ICMP utilities.

Advanced Tasks

1. Create a protocol wise packet statistics report using Wireshark
2. Investigate abnormal traffic patterns and identify suspicious packets.